

Football Predictions API Documentation

Base URL

```
http://127.0.0.1:8000/api/v1
```

Response format

Most endpoints return a consistent JSON wrapper:

```
{
  "success": true,
  "message": "Descriptive message",
  "data": {}
}
```

DELETE endpoints usually return **204 No Content** with an empty body.

Health Check

GET `/health`

Checks that the API is running.

Response

```
{
  "status": "ok",
  "app": "Football Predictions API"
}
```

Teams

GET `/teams`

Returns all teams.

Parameters None.

Response

```
{
  "success": true,
  "message": "Teams retrieved",
  "data": [
    {
      "id": 1,
      "name": "Roma",
      "country": "Italy",
      "coach": "Daniele De Rossi",
      "founded_year": 1927,
      "uefa_coefficient": 78.0
    }
  ]
}
```

GET /teams/{team_id}

Returns one team by ID.

Path parameters - team_id (integer, required): Team ID.

Response

```
{
  "success": true,
  "message": "Team retrieved",
  "data": {
    "id": 1,
    "name": "Roma",
    "country": "Italy",
    "coach": "Daniele De Rossi",
    "founded_year": 1927,
    "uefa_coefficient": 78.0
  }
}
```

POST /teams

Creates a team.

Request body

```
{
  "name": "Roma",
  "country": "Italy",
  "coach": "Daniele De Rossi",
  "founded_year": 1927,
  "uefa_coefficient": 78.0
}
```

Body fields - `name` (string, required) - `country` (string, required) - `coach` (string, optional) - `founded_year` (integer, optional) - `uefa_coefficient` (number, optional, default 0.0)

Response

```
{
  "success": true,
  "message": "Team created",
  "data": {
    "id": 1,
    "name": "Roma",
    "country": "Italy",
    "coach": "Daniele De Rossi",
    "founded_year": 1927,
    "uefa_coefficient": 78.0
  }
}
```

PUT /teams/{team_id}

Updates an existing team.

Path parameters - `team_id` (integer, required)

Request body Any subset of:

```
{
  "name": "Roma",
  "country": "Italy",
  "coach": "Daniele De Rossi",
  "founded_year": 1927,
  "uefa_coefficient": 80.0
}
```

DELETE /teams/{team_id}

Deletes a team.

Path parameters - `team_id` (`integer`, required)

Response - Status: `204 No Content`

Players

GET `/players`

Returns all players.

Parameters None.

Response

```
{
  "success": true,
  "message": "Players retrieved",
  "data": [
    {
      "id": 1,
      "name": "Paulo Dybala",
      "position": "Forward",
      "nationality": "Argentina",
      "age": 30,
      "goals": 3,
      "assists": 2,
      "rating": 7.7,
      "team_id": 1
    }
  ]
}
```

GET `/players/{player_id}`

Returns one player by ID.

Path parameters - `player_id` (`integer`, required)

POST `/players`

Creates a player.

Request body

```
{
  "name": "Paulo Dybala",
  "position": "Forward",
```

```

    "nationality": "Argentina",
    "age": 30,
    "goals": 3,
    "assists": 2,
    "rating": 7.7,
    "team_id": 1
  }

```

Body fields - `name` (string, required) - `position` (string, required) - `nationality` (string, required) - `age` (integer, required) - `goals` (integer, optional, default 0) - `assists` (integer, optional, default 0) - `rating` (number, optional, default 0.0) - `team_id` (integer, required)

PUT `/players/{player_id}`

Updates an existing player.

Path parameters - `player_id` (integer, required)

Request body Any subset of the player fields.

DELETE `/players/{player_id}`

Deletes a player.

Path parameters - `player_id` (integer, required)

Response - Status: 204 No Content

Matches

GET `/matches`

Returns all matches.

Parameters None.

Response

```

{
  "success": true,
  "message": "Matches retrieved",
  "data": [
    {
      "id": 1,
      "home_team_id": 1,
      "away_team_id": 2,

```

```

    "round_name": "Round of 16",
    "leg_number": 1,
    "bracket_slot": "R16-7",
    "match_date": "2026-03-12T20:00:00",
    "venue": "Stadio Olimpico",
    "is_neutral_venue": false,
    "home_score": null,
    "away_score": null,
    "status": "scheduled",
    "winner_team_id": null
  }
]
}

```

GET /matches/{match_id}

Returns one match by ID.

Path parameters - `match_id` (integer, required)

POST /matches

Creates a match.

Request body

```

{
  "home_team_id": 1,
  "away_team_id": 2,
  "round_name": "Round of 16",
  "leg_number": 1,
  "bracket_slot": "R16-7",
  "match_date": "2026-03-12T20:00:00",
  "venue": "Stadio Olimpico",
  "is_neutral_venue": false,
  "home_score": null,
  "away_score": null,
  "status": "scheduled",
  "winner_team_id": null
}

```

Body fields - `home_team_id` (integer, required) - `away_team_id` (integer, required) - `round_name` (string, required) - `leg_number` (integer, optional) - `bracket_slot` (string, optional) - `match_date` (datetime, required, ISO 8601) - `venue` (string, required) - `is_neutral_venue` (boolean, optional, default false) - `home_score` (integer, optional) - `away_score` (integer, optional) - `status` (string, optional, default scheduled) - `winner_team_id` (integer, optional)

PUT /matches/{match_id}

Updates an existing match.

Path parameters - match_id (integer, required)

Request body Any subset of the match fields.

DELETE /matches/{match_id}

Deletes a match.

Path parameters - match_id (integer, required)

Response - Status: 204 No Content

Team Form

GET /team-form

Returns all team form snapshots.

Parameters None.

Response

```
{
  "success": true,
  "message": "Team form snapshots retrieved",
  "data": [
    {
      "id": 1,
      "team_id": 1,
      "as_of_date": "2026-03-12T09:00:00",
      "last_5_wins": 4,
      "last_5_draws": 0,
      "last_5_losses": 1,
      "last_5_goals_for": 11,
      "last_5_goals_against": 5,
      "home_form_rating": 8.1,
      "away_form_rating": 7.5
    }
  ]
}
```

GET /team-form/{form_id}

Returns one team form snapshot by ID.

Path parameters - `form_id` (integer, required)

POST /team-form

Creates a team form snapshot.

Request body

```
{
  "team_id": 1,
  "as_of_date": "2026-03-12T09:00:00",
  "last_5_wins": 4,
  "last_5_draws": 0,
  "last_5_losses": 1,
  "last_5_goals_for": 11,
  "last_5_goals_against": 5,
  "home_form_rating": 8.1,
  "away_form_rating": 7.5
}
```

Body fields - `team_id` (integer, required) - `as_of_date` (datetime, required) - `last_5_wins` (integer, required) - `last_5_draws` (integer, required) - `last_5_losses` (integer, required) - `last_5_goals_for` (integer, required) - `last_5_goals_against` (integer, required) - `home_form_rating` (number, required) - `away_form_rating` (number, required)

PUT /team-form/{form_id}

Updates an existing team form record.

Path parameters - `form_id` (integer, required)

Request body Any subset of the form fields.

DELETE /team-form/{form_id}

Deletes a team form record.

Path parameters - `form_id` (integer, required)

Response - Status: 204 No Content

Match Predictions

GET /match-predictions

Returns all stored match predictions.

Parameters None.

Response

```
{
  "success": true,
  "message": "Match predictions retrieved",
  "data": [
    {
      "id": 1,
      "match_id": 1,
      "model_name": "poisson_form_model",
      "model_version": "1.0.0",
      "generated_at": "2026-03-12T12:00:00Z",
      "home_win_probability": 0.42,
      "draw_probability": 0.27,
      "away_win_probability": 0.31,
      "predicted_home_goals": 1.48,
      "predicted_away_goals": 1.21,
      "confidence_score": 0.42
    }
  ]
}
```

GET /match-predictions/{prediction_id}

Returns one prediction by ID.

Path parameters - prediction_id (integer, required)

POST /match-predictions

Creates a prediction manually.

Request body

```
{
  "match_id": 1,
  "model_name": "poisson_form_model",
  "model_version": "1.0.0",
  "generated_at": "2026-03-12T12:00:00Z",
  "home_win_probability": 0.42,
```

```
"draw_probability": 0.27,  
"away_win_probability": 0.31,  
"predicted_home_goals": 1.48,  
"predicted_away_goals": 1.21,  
"confidence_score": 0.42  
}
```

Body fields - `match_id` (integer, required) - `model_name` (string, required) - `model_version` (string, required) - `generated_at` (datetime, required) - `home_win_probability` (number, required, 0-1) - `draw_probability` (number, required, 0-1) - `away_win_probability` (number, required, 0-1) - `predicted_home_goals` (number, optional) - `predicted_away_goals` (number, optional) - `confidence_score` (number, optional, 0-1)

PUT `/match-predictions/{prediction_id}`

Updates a stored prediction.

Path parameters - `prediction_id` (integer, required)

DELETE `/match-predictions/{prediction_id}`

Deletes a prediction.

Path parameters - `prediction_id` (integer, required)

Response - Status: 204 No Content

Prediction Generation

POST `/predictions/generate`

Automatically generates predictions for matches that do not already have one.

Parameters None.

Response

```
{  
  "success": true,  
  "message": "Predictions generated",  
  "count": 16  
}
```

Tournament Simulation

GET /tournament/simulate-once

Runs one full tournament simulation from the Round of 16 to the final.

Parameters None.

Response

```
{
  "success": true,
  "data": {
    "round_of_16": [],
    "quarter_finals": [],
    "semi_finals": [],
    "final": {
      "match": {
        "home_team": "Roma",
        "away_team": "Porto",
        "home_goals": 2,
        "away_goals": 1
      },
      "winner_team_id": 1,
      "winner_team_name": "Roma"
    },
    "champion": "Roma"
  }
}
```

GET /tournament/simulate-many

Runs multiple tournament simulations and returns estimated title probabilities.

Query parameters - runs (integer, optional, default 1000): Number of simulations to run.

Example

```
GET /tournament/simulate-many?runs=1000
```

Response

```
{
  "success": true,
  "data": {
    "runs": 1000,
    "champion_probabilities": [
```

```
{
  {
    "team": "Roma",
    "titles": 186,
    "probability": 0.186
  },
  {
    "team": "Porto",
    "titles": 173,
    "probability": 0.173
  }
]
}
```

Analytics

GET /analytics/leaderboard

Returns a leaderboard calculated from completed matches.

Parameters None.

Response

```
{
  "success": true,
  "message": "Leaderboard generated",
  "data": [
    {
      "team_id": 1,
      "team_name": "Roma",
      "wins": 3,
      "draws": 1,
      "losses": 1,
      "goals_for": 10,
      "goals_against": 5,
      "points": 10
    }
  ]
}
```

GET /analytics/players/summary

Returns player summaries.

Parameters None.

Response

```
{
  "success": true,
  "message": "Player summaries generated",
  "data": [
    {
      "player_id": 1,
      "player_name": "Paulo Dybala",
      "team_name": "Roma",
      "goals": 3,
      "assists": 2,
      "rating": 7.7
    }
  ]
}
```

GET /analytics/predictions/summary

Returns prediction summaries including team names and model metadata.

Parameters None.

Response

```
{
  "success": true,
  "message": "Prediction summaries generated",
  "data": [
    {
      "prediction_id": 1,
      "match_id": 1,
      "model_name": "poisson_form_model",
      "model_version": "1.0.0",
      "home_team": "Roma",
      "away_team": "Porto",
      "home_win_probability": 0.42,
      "draw_probability": 0.27,
      "away_win_probability": 0.31,
      "confidence_score": 0.42
    }
  ]
}
```

Common status codes

- 200 OK – Request successful
- 201 Created – Resource created successfully
- 204 No Content – Resource deleted successfully
- 400 Bad Request – Validation error or invalid foreign key reference
- 404 Not Found – Resource does not exist
- 409 Conflict – Duplicate resource, such as an existing team name
- 422 Unprocessable Entity – Request body validation failed

Notes

- Dates should be sent in ISO 8601 format.
- Predictions depend on match and team form data already being present.
- Tournament simulation depends on exactly eight Round of 16 ties existing in the database.
- Interactive documentation is available at `/docs` when the API is running.